Applied Unsupervised Learning in Python

Today's goal • 3/3 items

Module 1
Basic Unsupervised Learning Methods

Welcome to the Course

- Welcome to Applied Unsupervised Learning in Python
  Video • 5 min
- MADSwPy Certificate Roadmap
  Reading • 5 min
- Course Syllabus
  Reading • 10 min
- Additional Resources
  Reading • 10 min
- Meet Your Fellow Learners
  Discussion Prompt • 15 min
- Help Us Learn About You
  Reading • 5 min

Lesson 1: Methods for Dimensionality Reduction

- Dimensionality Reduction: A Brief Introduction
  Video • 16 min
- Dimensionality Reduction with Feature Selection: Information Gain
  Video • 15 min
- Time to Practice: Dimensionality Reduction
  Graded Assignment • Grade: 100%
- Dimensionality Reduction with Feature Selection: Principal Component Analysis (PCA) Explained
  Video • 20 min
- Visualizing PCA Results: Foundations
  Video • 8 min
- Visualizing PCA Results: Biplots and Variance Plots
  Video • 17 min
- Ten Quick Tips for Effective Dimensionality Reduction
  Reading • 45 min
- Time to Practice: Principal Component Analysis (PCA)
  Graded Assignment • Grade: 100%
- Singular Value Decomposition (SVD)
  Video • 21 min
- Applications of SVD in Data Science
  Video • 13 min
- Time to Practice: Singular Value Decomposition (SVD)
  Graded Assignment • Grade: 100%
- Manifold Learning: Multidimensional Scaling (Part 1)
  Video • 17 min
- Manifold Learning: Multidimensional Scaling (Part 2)
  Video • 11 min
- Time to Practice: Manifold Learning (Multidimensional scaling: Parts 1 & 2)
  Graded Assignment • Grade: 100%
- Manifold Learning: t-Distributed Stochastic Neighbor Embedding (t-SNE)
  Video • 13 min
- Manifold Learning: Uniform Manifold Approximation and Projection (UMAP)
  Video • 6 min
- Time to Practice: Manifold Learning (t-SNE, and UMAP)
  Graded Assignment • Grade: 100%

Lesson 2: Density Estimation

- Density Estimation Part 1: Probability Density Functions
  Video • 12 min
- Density Estimation Part 1: Parametric vs. Non-Parametric Density Estimator
  Video • 13 min
- Time to Practice: Density Estimation Methods (Part 1)
  Graded Assignment • Grade: 100%
- Density Estimation Part 2: Local Density Estimators
  Video • 11 min
- Density Estimation Part 2: Kernel Density Estimators
  Video • 15 min
- Density Estimation Part 2: Evaluating Density Estimators
  Video • 4 min

Module 1 Optional Readings & Resources

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- **Textbook Chapter:** Introduction to Machine Learning with Python Section 3.1: Types of Unsupervised Learning; Section 3.2: Challenges in Unsupervised Learning; Section 3.4: Dimensionality Reduction, Feature Extraction, and Manifold Learning.
- **Article:** How to Use t-SNE Effectively This article by engineers from Google uses effective interactive demos to explain how to adjust key parameters of t-SNE to get the most out of this visualization method.
- **Advanced Textbook Chapter:** Elements of Statistical Learning, Chapter 14 This chapter goes into much more technical depth on some areas of unsupervised learning, especially dimensionality reduction and clustering.
- **Article:** Principal Components Analysis: A Review and Recent Developments A clearly written explanation of PCA for those who'd like to go into slightly more technical/statistical depth or simply view the topic from another perspective.

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